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Transmission of Signal Through Continuous Time LTI Systems

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Answer key	at the end

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Marks 1 12 questions

Q1. GATE ECE 2022

Let $x_1(t) = e^{-t}\,u(t)$ and $x_2(t) = u(t) - u(t-2)$, where $u(\,.\,)$ denotes the unit step function. If $y(t)$ denotes the convolution of $x_1(t)$ and $x_2(t)$, then $\lim_{t\to\infty} y(t)$ = _____ (rounded off to one decimal place).

Open solution

Q2. GATE ECE 2016 Set 1

A continuous-time sinusoid of frequency 33 Hz is multiplied with a periodic Dirac impulse train of frequency 46 Hz. The resulting signal is passed through an ideal analog low-pass filter with a cutoff frequency of 23Hz. The fundamental frequency (in Hz) of the output is _____.

Open solution

Q3. GATE ECE 2014 Set 4

A real - values signal x(t) limited to the frequency band $|f| \le \frac{W}{2}$ is passed through a linear time invariant system whose frequency response is

$$H(f) = \begin{cases} e^{-j4\pi f}, & |f| \le \frac{W}{2} \\ 0, & |f| > \frac{W}{2} \end{cases}$$

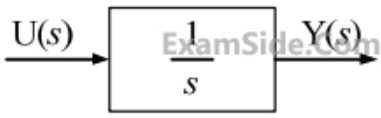
The output of the system is

(A) x(t + 4) (B) x(t - 4) (C) x(t + 2) (D) x(t - 2)

Open solution

Q4. GATE ECE 2013

Assuming zero initial condition, the response y (t) of the system given below to a unit step input u(t) is



(A) u (t) (B) tu (t) (C) $\frac{t^2}{2}(t)$ (D) $e^{-t}u(t)$

Open solution

Q5. GATE ECE 2013

Let g(t) = $e^{-\pi t^2}$, and h(t) is a filter matched to g(t). If g(t) is applied as input to h(t), then the Fourier transform of the output is

(A) $e^{-\pi f^2}$ (B) $e^{-\pi f^2/2}$ (C) $e^{-\pi|f|}$ (D) $e^{-2\pi f^2}$

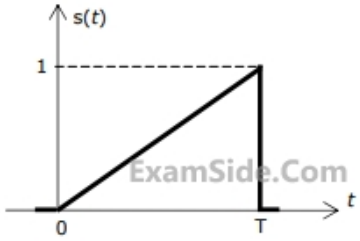
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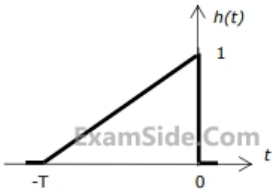
Q6. GATE ECE 2010

Open solution

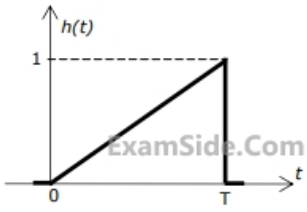
Consider the pulse shape s(t) as shown. The impulse response h(t) of the filter matched to this pulse is



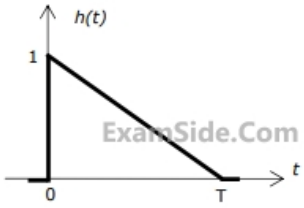
(A)



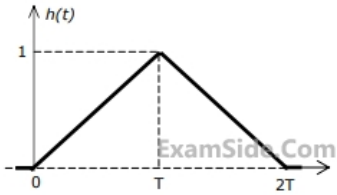
(B)



(C)



(D)



Q7. GATE ECE 2010

Open solution

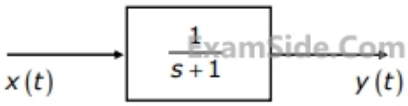
A system with the transfer function $\frac{Y(s)}{X(s)} = \frac{s}{s+p}$ has an output $y(t) = \cos\left(2t - \frac{\pi}{3}\right)$ for the input signal $x(t) = p\cos\left(2t - \frac{\pi}{2}\right)$. Then, the system parameter 'p' is

(A) $\sqrt{3}$ (B) $\frac{2}{\sqrt{3}}$ (C) 1 (D) $\frac{\sqrt{3}}{2}$

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Q8. GATE ECE 2006

In the system shown below,
x(t) = (sint)u(t). In steady-state, the response y(t) will be



(A) $\frac{1}{\sqrt{2}}\sin\left(t-\frac{\pi}{4}\right)$
(B) $\frac{1}{\sqrt{2}}\sin\left(t+\frac{\pi}{4}\right)$
(C) $\frac{1}{\sqrt{2}}e^{-t}\sin(t)$
(D) $\sin(t)-\cos(t)$

Open solution

Q9. GATE ECE 2006

A low-pass filter having a frequency response $H(j\omega) = A(\omega)e^{j\Phi(\omega)}$, does not product any phase distortion if

(A) $A(\omega)=C\omega^2,\ \phi(\omega)=K\omega^3$
(B) $A(\omega)=C\omega^2,\ \phi(\omega)=K\omega$
(C) $A(\omega)=C\omega,\ \phi(\omega)=K\omega^2$
(D) $A(\omega)=C,\ \phi(\omega)=K\omega^{-1}$

Open solution

Q10. GATE ECE 2002

A linear phase channel with phase delay τ_p and group delay τ_g must have

(A) $\tau_p=\tau_g=$ constant
(B) $\tau_p \infty\ f\ and\ \tau_g \infty\ f$
(C) τ_p = constant and $\tau_g \infty\ f$
(D) $\tau_p \infty\ f\ and\ \tau_g$ =constant ($f$denotes frequency)

Open solution

Q11. GATE ECE 1999

The input to a channel is a band pass signal. It is obtained by linearly modulating a sinusoidal carrier with a signal-tone signal. The output of the channel due to this input is given by y(t) = (1/100) $\cos(100t-10^{-6})\ \cos(10^6t-1.56)$. The group delay (t_g) and the phase delay (t_p), in seconds, of the channel are

(A) $t_g=10^{-6},\ t_p=1.56$
(B) $t_g=1.56,\ t_p=10^{-6}$
(C) $t_g=\ 10^{-8},\ t_p=1.56\times10^{-6}$
(D) $t_g=10^{-8},\ t_p=1.56$

Open solution

Q12. GATE ECE 1996

A rectangular pulse of duration T is applied to a filter matched to this input. The output of the filter is a

(A) rectangular pulse of duration T.
(B) rectangular pulse of duration 2T.
(C) triangular pulse.
(D) sine function.

Open solution

Marks 2 17 questions

Q13. GATE ECE 2023

Let $x(t)=10\cos(10.5Wt)$ be passed through an LTI system having impulse response $h(t)=\pi\left(\frac{\sin Wt}{\pi t}\right)^2\cos 10Wt$. The output of the system is

(A) $\left(\frac{15W}{4}\right)\cos(10.5Wt)$
(B) $\left(\frac{15W}{2}\right)\cos(10.5Wt)$
(C) $\left(\frac{15W}{8}\right)\cos(10.5Wt)$
(D) $(15W)\cos(10.5Wt)$

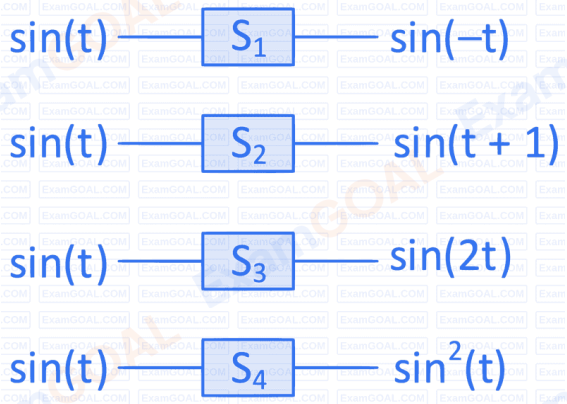
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Q14. GATE ECE 2022

The outputs of four systems (S_1, S_2, S_3 and S_4) corresponding to the input signal sin(t), for all time t, are shown in the figure.

Based on the given information, which of the four systems is/are definitely NOT LTI (linear and time-invariant)?



(A) S_1 (B) S_2 (C) S_3 (D) S_4

Open solution

Q15. GATE ECE 2016 Set 2

A continuous-time filter with transfer function $H(S) = \frac{2s+6}{s^2+6s+8}$ is converted to a discrete time filter with transfer function $G(Z) = \frac{2z^2-0.5032\,z}{z^2-0.5032z+k}$ so that the impulse response of the continuous-time filter, sampled at 2 Hz, is identical at the sampling instants to the impulse response of the discrete time filter, The value of k is _____.

Open solution

Q16. GATE ECE 2016 Set 1

A first-order low-pass filter of time constant T is excited with different input signals (with zero initial conditions up to t = 0). Match the excitation signals X, Y, Z with the corresponding time responses for $t \geq 0$:

X: Impulse P: 1 $-e^{-t/T}$

Y: Unit step Q: t $-T(1-e^{-t/T})$

Z: Ramp R: $e^{-t/T}$

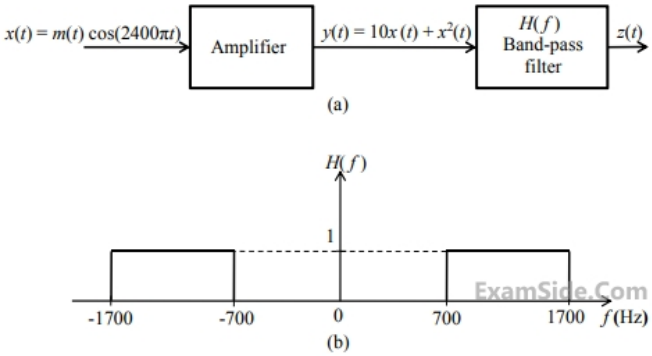
(A) $X \to R,\ Y \to Q,\ Z \to P$ (B) $X \to Q,\ Y \to P,\ Z \to R$

(C) $X \to R,\ Y \to P,\ Z \to Q$ (D) $X \to P,\ Y \to R,\ Z \to Q$

Open solution

Q17. GATE ECE 2015 Set 1

In the system shown in Figure (a), m(t) is a low-pass signal with bandwidth W Hz. The frequency response of the band-pass filter H(f) is shown in Figure (b). If it is desired that the output signal z(t) = 10x(t), the maximum value of W (in Hz) should be strictly less than _____



Open solution

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Q18. GATE ECE 2007

The 3 - dB bandwidth of the low - pass signal e^{-1} u(t), where u(t) is the unit step function, is given by

(A) $\frac{1}{2\pi} Hz$ (B) $\frac{1}{2\pi}\sqrt{\sqrt{2}-1}\,Hz$

(C) ∞ (D) 1 Hz

Open solution

Q19. GATE ECE 2004

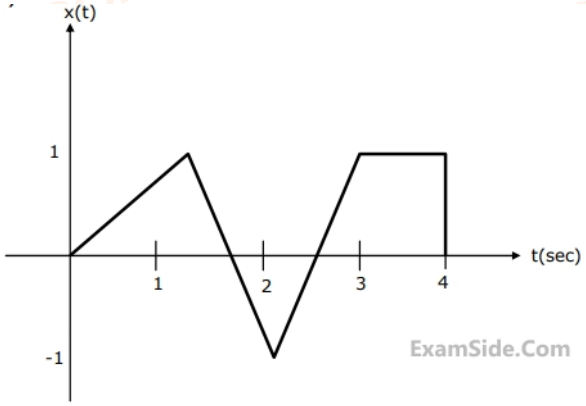
A system has poles at 0.01 Hz, 1 Hz and 80 Hz; zeros at 5 Hz, 100 Hz and 200 Hz. The approximate phase of the system response at 20 Hz is

(A) -90° (B) 0° (C) 90° (D) -180°

Open solution

Q20. GATE ECE 2004

Consider the signal x(t) shown in Fig. Let h(t) denote the impulse response of the filter matched to x(t), with h(t) being non-zero only in the interval 0 to 4 sec. The slope of h(t) in the interval 3 < t < 4 sec is



(A) $1/2\,\sec^{-1}$ (B) $-1\,\sec^{-1}$ (C) $-1/2\,\sec^{-1}$ (D) $1\,\sec^{-1}$

Open solution

Q21. GATE ECE 2003

The system under consideration is an RC low -pass filter (RC-LPF) with R = 1.0 $k\Omega$ and C = 1.0 μF. Let H(t) denote the frequency response of the RC-LPF. Let f_1 be the highest frequency such that $0 \le |f| \le f_1$, $\frac{|H(f_1)|}{H(0)} \ge 0.95$. Then f_1 (in Hz) is

(A) 327.8 (B) 163.9 (C) 52.2 (D) 104.4

Open solution

Q22. GATE ECE 2003

The system under consideration is an RC low -pass filter (RC-LPF) with R = 1.0 $k\Omega$ and C = 1.0 μF. Let t_g (f) be the group delay function of the given RC-LPF and f_2 = 100 Hz. Then $t_g(f_2)$ in ms, is

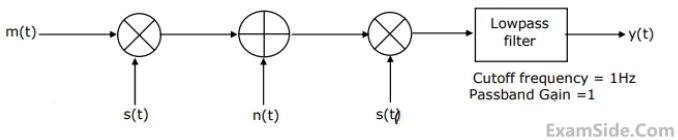
(A) 0.717 (B) 7.17 (C) 71.7 (D) 4.505

Open solution

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Q23. GATE ECE 2002

In Fig. m(t) = $= \frac{2\sin 2\pi t}{t}$, $s(t) = \cos\ 200\pi t\ and n(t) = \frac{\sin 199\pi t}{t}$.
The output y(t) will be



Open solution

(A) $\frac{\sin 2\pi\, t}{t}$
(B) $\frac{\sin 2\pi\, t}{t} + \frac{\sin \pi\, t}{t}\cos\ 3\pi t$
(C) $\frac{\sin 2\pi\, t}{t} + \frac{\sin 0.5\,\pi\, t}{t}\cos\ 1.5\pi t$
(D) $\frac{\sin 2\pi\, t}{t} + \frac{\sin \pi\, t}{t}\cos\ 0.75\pi t$

Q24. GATE ECE 2000

A system has a phase response given by $\phi\,(\omega)$ where ω is the angular frequency. The phase delay and group delay at ω = ω_0 are respectively given by

Open solution

(A) $-\frac{\phi(\omega_0)}{\omega_0}, -\frac{d\phi(\omega)}{d\omega}\,|\omega = \omega_0$
(B) $\phi(\omega_0), -\frac{d^2\phi(\omega)}{d\omega^2}\,|\omega = \omega_0$
(C) $\frac{\omega_0}{\phi(\omega_0)}, -\frac{d\phi(\omega)}{d\omega}\,|\omega = \omega_0$
(D) $\omega_0\,\phi\,(\omega_0)\,,\ \int_{-\infty}^{\omega_0}\phi(\lambda)\,d\,\lambda$

Q25. GATE ECE 1999

The input to a matched filter is given by $s(t) = \begin{cases} 10\sin(2\pi \times 10^6 t), & 0 < |t| < 10^{-4}\,\text{sec} \\ 0 & Otherwise \end{cases}$

The peak amplitude of the filter output is

Open solution

(A) 10 Volts (B) 5 Volts (C) 10 millivolts (D) 5 millivolts

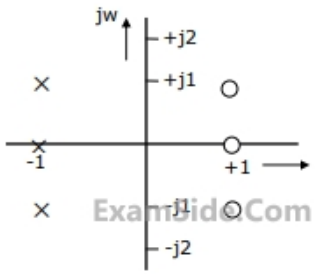
Q26. GATE ECE 1993

Sketch the waveform (with properly marked axes) at the output of a matched filter matched for a signal s(t), of duration T, given by $s(t) = \begin{cases} A & for \quad 0 \le t < \frac{2}{3}T \\ 0 & for \quad \frac{2}{3}T \le t < T \end{cases}$

Open solution

Q27. GATE ECE 1991

The pole-zero pattern of a certain filter is shown in the Fig. The filter must be of the following type.



Open solution

(A) low pass (B) high pass (C) all pass (D) band pass

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Q28. GATE ECE 1990

The magnitude and phase transfer functions for a distortionless filter should respectively be:

(A) Magnitude = Linear, Phase = Constant
(B) Magnitude = Constant, Phase = Constant
(C) Magnitude =Constant , Phase = Linear
(D) Magnitude = Linear, Phase = Linear

Open solution

Q29. GATE ECE 1988

Specify the filter type if its voltage transfer function H(s) is given by
H(s) = $\frac{K(s^2+\omega_0{}^2)}{s^2+(\omega_0/Q)s+\omega_0{}^2}$

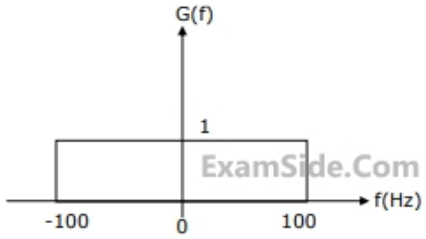
(A) all pass filter (B) low pass filter (C) band pass filter (D) notch filter

Open solution

Marks 5 5 questions

Q30. GATE ECE 2001

A base band signal g(t) band limited to 100 Hz modulates a carrier of frequency f_0 Hz. The modulated signal g(t) $\cos 2\pi f_0 t$ is transmitted over a channel whose input x and output y are related by $y = 2x + x^2$. The spectrum of g(t) is shown in Fig. Sketch the spectrum of the transmitted signal and the spectrum of the received signal.



Open solution

Q31. GATE ECE 1996

A signal 3 sin $(\pi\ f_0 t) + 5\ \cos\ (3\pi\ f_0 t)$ is applied to an RC low pass filter of 3 dB cutoff frequency f_0. Determine and plot the output power spectrum and aslo calculate the total input and output normalized power.

Open solution

Q32. GATE ECE 1996

An input signal A exp $(-\alpha\, t)$ u(t) with $\alpha > 0$ is applied to a causal filter, the impulse response of which is A exp $(-\alpha\ t)$. Determine the filter output, sketch it as a function of time and lable the important points.

Open solution

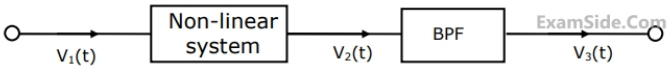
Q33. GATE ECE 1994

A signal, f(t) = e^{-at} u(t), where u(t) is the unit step function, is applied to the input of a low-pass filter having $|H(\omega)| = \frac{b}{\sqrt{\omega^2+b^2}}$.
Calculate the value of the ratio, $\frac{a}{b}$, for which 50% of the input signal energy is transferred to the output.

Open solution

Q34. GATE ECE 1993

Obtain an expression for the signal in figure, for the signal $v_3(t)$ in Fig for $v_1(t) = 100\cos(2000\pi t) + 4\sin(200\pi t)$.
Assume that $v_2(t)$=$v_1(t)$+0.1 $v_1^2(t)$ and that the BPF is an ideal unity gain filter with pass band from 800 Hz to 1200 Hz.



Open solution

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Answer Key

Marks 1

1. **0**	2. **13**	3. **D**	4. **B**	5. **D**
6. **C**	7. **B**	8. **A**	9. **B**	10. **A**
11. **C**	12. **C**			

Marks 2

13. **A**	14. **C, D**	15. **0.05**	16. **C**	17. **350**
18. **A**	19. **C**	20. **B**	21. **C**	22. **A**
23. **C**	24. **A**	25. **D**	26. **solve it.**	27. **C**
28. **C**	29. **D**			

Marks 5

30. **solve it.**	31. **solve it.**	32. **solve it.**	33. $a/b = 1$	34. $v_3(t) = 10\cos(2000\pi t) + 8\cos(2$

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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